

FINITE SYMBOLIC MECHANICS

Finite Representation and Interactional Extent

From the prototype metre to the compressed parsec

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Working monograph • 23 September 2026

Abstract

A short inscription can stand for a long chain of measured and inferred relations. The expression “1 m” may refer to an interval set by comparison with a physical standard, a length realised through contemporary metrology, or a spatial array described at a stipulated resolution. These are different trajectories compressed into similar symbols. This monograph constructs a conditional account of their cost under the licence of Finite Symbolic Mechanics (FSM). It distinguishes the cost of an inscription, the information needed to specify a length, the cost of an explicitly resolved representation, and the resources expended in an actual measurement. It introduces a resolution dependent ladder from millimetre to megaparsec, keeps the Alphonic distinction scale separate from the Greene–Pascal condition for a usable trajectory, and locates comparison remainders at the bridges between registrations and models. Finally it gives a research programme for bringing this account to classical physics, beginning with local optical measurements and proceeding carefully to spectroscopy and astronomical distance inference. The interval counts developed here are conditional properties of a representation. Whether any associated structure changes the propagation of light or the outcomes of other interactions remains an empirical question.

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Chapter 1

A licence for measured expressions

A Theory of Finite Representation, Part I established the conditions under which a finite symbol can acquire scientific use. *Registration of Extent* then followed an extended physical interaction towards a stable registration and its symbolic expression. The present monograph takes up the compression at that crossing. Once an inscription such as “1 m” is licensed to stand for an extent, what must be retained to reconstruct its meaning at a declared resolution, and which of those representational costs, if any, enter a physical model?

A physical equation can be manipulated as a mathematical expression and used as a claim about independently realised quantities. Those activities are connected, but their permissions differ. In classical algebra the statement $F = ma$ identifies formal expressions once its terms and operations have been specified. In an experiment, a registered force, a registered mass and a registered acceleration are reached by different paths through instruments, comparisons, records and calibrations. Under FSM a physical term is admitted through such a finite path; the mathematical identity does not by itself establish interactional identity among the registrations.

This distinction explains the historical role of k_{ma} in the earlier Finite Mechanics expression $F \mid ma + k_{ma}$. The vertical bar is used here for a licensed comparison or interactional correspondence, with the licence stated in context; it is not an alternative spelling of an algebraic equals sign. The older k_{ma} made a remainder visible when trajectories were brought together. It did not thereby establish a new force. The most general initial expression is a comparison record

$$\mathcal{R}_F \mid_{\mathcal{L}} \Phi(\mathcal{R}_m, \mathcal{R}_a), \quad \kappa_{\mathcal{L}}(\mathcal{R}_F, \Phi(\mathcal{R}_m, \mathcal{R}_a)), \quad (1.1)$$

where $\mathcal{R}_i$ denotes a finite registration, Φ a declared construction, $\mathcal{L}$ the licence for comparison and $\kappa_{\mathcal{L}}$ whatever remains unresolved by that comparison. The notation κ is a record of a relation; its form might be an interval, pattern, covariance, unmatched provenance or measured difference. Writing an additive k_{ma} is a further decision justified only when the remainder can be given a compatible magnitude and operation. Nor does the impossibility of a zero-uncertainty physical registration alone entail an additive correction.

Classical set theory licenses exact formal objects; it does not itself make an assertion about an instrument or a mark on paper. The relevant bridge occurs when measured outcomes are represented by those objects. FSM asks which exact properties of a formal surrogate have been established for the particular registration, and which have merely been carried across under a modelling licence. That question is constructive: it permits the mathematical calculation to continue while preserving a record of its translation into a physical claim.

Chapter 2

The metre as a trajectory

The platinum–iridium international prototype offers a tangible first case. The international metre was sanctioned in 1889 in relation to the prototype; the later operational specification located the length between the axes of two marks, with specified temperature, pressure and support conditions. The bar itself was never a sequence of perfect units laid end to end. To reproduce its length one compared marks through optics, thermal conditions, apparatus and records. A finite registration was translated into an authorised unit by a chain of operations. In 1960 the definition changed to a specified radiation, and the present SI defines the metre through the fixed numerical value assigned to the vacuum speed of light together with the definition of the second. Each route licenses the compact symbol m while changing the provenance and operations by which a length is realised.¹

A claim about the cost of representing this metre therefore needs an explicit task. To record a measured length to a certain precision is one task. To print the unit name is another. To draw or store a spatially resolved record across the entire extent is a third. To reproduce the prototype by a laboratory comparison is a fourth. Their resources need not be equal. In particular, printing “1 m” does not execute 10^{11} physical operations merely because we choose, later, to imagine the length partitioned into 10 pm intervals.

Let an extent $L > 0$ be presented under a resolution rule $\Delta > 0$. A simple coverage count is

$$N_{\text{cov}}(L, \Delta) = \left\lceil \frac{L}{\Delta} \right\rceil. \quad (2.1)$$

The ceiling counts required intervals when a final partial interval still needs representation. For the exact endpoint grid one might count $N_{\text{cov}} + 1$ boundaries instead; the choice must be declared. Equation (2.1) asserts neither that space is granular nor that every interval has been separately measured. It counts places in one chosen representation. The resolution Δ describes a task or registration protocol, not an intrinsic distance at which physical space stops.

¹BIPM, *History of the metre*, <https://www.bipm.org/en/history-si/metre>; BIPM, *Metre*, <https://www.bipm.org/en/si-base-units/metre>; 7th CGPM (1927), Resolution 1, <https://www.bipm.org/en/committees/cg/cgpm/7-1927/resolution-1>.

Chapter 3

Four costs within one symbol

We can now prevent the word “cost” from doing several incompatible jobs. Let s be an inscription in a declared alphabet and encoding e . Its *surface cost* $C_{\text{surf}}(s; e)$ is its character or byte length. It is sensitive to typography, conventions and coding: “1 m”, “1 metre” and a binary value can designate corresponding lengths without having the same surface cost. The count alone carries no physical dimension of energy or force.

A second cost is the *scalar description cost*. Suppose a length known only to lie between 0 and L_{max} is reported in bins of width Δ . Without additional prior information there are $M = \lfloor L_{\text{max}}/\Delta \rfloor + 1$ possible bin labels, requiring at least $\lceil \log_2 M \rceil$ bits in a fixed-length binary index. This grows logarithmically with range and finer resolution. The format, units, metadata and uncertainty may require more, but the bit count does not ordinarily grow as L/Δ . A scalar coordinate and a complete spatial record are different objects.

A third cost is the *explicit resolved record*. If a protocol asks for a label at each of N_{cov} intervals and each label uses b bits, then a direct array uses approximately bN_{cov} bits, before metadata. This is a conditional storage estimate, not a universal lower bound: a uniform array can be compressed to a short generating rule. If each interval can vary independently among q distinguishable states and the task requires all states to be recoverable, there are q^N possible arrays and the worst-case discriminating information is $N \log_2 q$ bits. Correlated states and a known model reduce the required description. It is therefore essential to state *which differences must survive decompression*.

A fourth cost is the *physical realisation cost*: detector events, elapsed time, calibration work, heat, computation, transmitted energy or other resources under a particular experimental design. We denote it $C_{\text{phys}}(P)$ for protocol P , leaving its dimensions explicit. No constant conversion from number of representational intervals to joules, or from a written character to a physical event, has been established here. A useful accounting object is consequently a vector rather than one unqualified number:

$$\mathcal{C}(L, \Delta; P, e) = (C_{\text{surf}}, C_{\text{scalar}}, C_{\text{array}}, C_{\text{phys}})_{L, \Delta; P, e}. \quad (3.1)$$

A comparison or ratio is licensed only between like components under a shared task. For example, $\Gamma_{\text{array/surf}} = C_{\text{array}}/C_{\text{surf}}$ can illustrate compression after both costs have been expressed in bits and a specific encoding chosen. It is a ratio of descriptions, not a propagation law.

For a deliberately simple character account, the ASCII inscriptions “1 m” and “1 pc” use respectively three and four bytes, or 24 and 32 bits before metadata. If an explicit 10 pm grid is assigned one bit per interval, its nominal sizes are 10^{11} bits for a metre and 3.086×10^{27} bits for a parsec; dividing by those surface sizes yields about 4.17×10^9 and

9.64×10^{25} . These are illustrative compression ratios for the two stipulated formats. They do not say that either quantity requires an array, nor that the characters themselves carry the energy of such an array.

Chapter 4

Alphonic distinction and the Greene–Pascal condition

In the developing FSM vocabulary, the Alphonic scale α names a working scale for a finite distinguishable registration in a stated practice. Earlier discussion proposed 10–20 pm as an illustrative anchor associated with structural measurements. It is a provisional, protocol dependent premise of the examples here, not an established universal smallest feature of matter, measurement or language. In the numerical ladder below we choose $\Delta = 10$ pm or 20 pm by stipulation. A future account must identify the instrument, contrast threshold, uncertainty and rule that make such a distinction repeatable.

The Greene–Pascal condition concerns a different question: how much finite trajectory must be retained before a mark or chain can function as a usable symbol? Denote a particular criterion by $G_{\text{GP}}(P, H, U)$, where P is a protocol, H its provenance and U its uncertainty record. A mark may contain candidate Alphonic distinctions without reaching the threshold for a stable comparison, a readable inscription or a communicable measurement. Conversely a usable symbol may gather many registrations into a compact name. Thus α may parameterise the smallest selected distinction, while G_{GP} expresses an admissibility condition upon a trajectory; setting them equal would erase the distinction the theory needs.

One can propose a provisional weight c_{GP} per retained interval and write $C_{\text{grid}} = c_{\text{GP}}N_{\text{cov}}$. This is a stipulated cost model for a required explicit grid. It is not a deduction that every metre physically contains N_{cov} independent Greene–Pascal events. Such a deduction would need evidence that the intervals are independently resolvable and actually relevant to the interaction in question.

Chapter 5

The resolution ladder

At millimetre resolution, a metre requires 10^3 intervals in the coverage convention. At 10 pm, the stipulated number is 10^{11} ; at 20 pm it is 5×10^{10} . The inscription can remain “1 m” across those choices. This is the first practical meaning of a latent decompression cost: a fine explicit record demands a much larger representation although the compact label changes little.

Table 5.1: Conditional coverage counts. Lengths are nominal unit conversions; counts are not observations of independently occupied cells. Values are rounded where shown.

Extent L	L in metres	N_{cov} at 1 mm	N_{cov} at 10 pm
1 mm	10^{-3}	1	10^8
1 m	1	10^3	10^{11}
1 km	10^3	10^6	10^{14}
1 au	$1.495978707 \times 10^{11}$	1.496×10^{14}	1.496×10^{22}
1 pc	$3.08567758 \times 10^{16}$	3.086×10^{19}	3.086×10^{27}
1 Mpc	$3.08567758 \times 10^{22}$	3.086×10^{25}	3.086×10^{33}

The astronomical unit has an exact adopted value of 149 597 870 700 m. The conventional parsec is $648\,000/\pi$ au, approximately $3.08567758 \times 10^{16}$ m; multiplying by 10^6 gives a megaparsec. These nominal conversions set the ladder but do not imply that every astronomical baseline has been directly measured to 10 pm.¹

The count changes in direct proportion to $1/\Delta$ away from ceiling effects. For a fixed L , doubling the stipulated interval from 10 to 20 pm halves the count. For a fixed Δ , multiplying extent by 10^6 multiplies the count by 10^6 . A 10 pm grid across one parsec contains approximately 3.086×10^{27} interval positions, which is approximately 3.086×10^{16} times the count across one metre. It would be a mistake to infer that a numerical specification of 1 pc requires that many digits; describing the upper endpoint of a scalar grid up to one parsec needs only about 92 binary positions. The compression hides a possible explicit reconstruction, not a proven physical expenditure made by the act of saying “parsec”.

¹IAU, *Measuring the Universe*, <https://iauarchive.eso.org/public/themes/measuring/>; IAU, *Recommendations concerning Units*, https://iauarchive.eso.org/publications/proceedings_rules/units/; IAU 2015 Resolution B2, https://www.iau.org/static/resolutions/IAU2015_English.pdf.

Chapter 6

Finite interactional extents and overlapping registrations

The elementary comparison regions of an instrument may overlap. If a detector has a spatial response kernel $h_i(x)$ for its i th sampling location, successive registrations need not observe isolated disjoint intervals. Let r_i be the recorded response and let $\Sigma_{ij} = \text{Cov}(r_i, r_j)$ denote their covariance under a specified ensemble and protocol. A large nominal N_{cov} may coexist with a much smaller number of recoverable independent distinctions because the response kernels overlap, because the signal is smooth, or because uncertainty prevents stable separation. A single scalar “overlap” parameter ω can be useful in a controlled model, but a response map and covariance matrix preserve more of the structure.

For an initial ledger write $\rho_P(x)$ for the density of required registrations per unit extent, $\omega_P(x)$ for an overlap descriptor where defined, U_P for the uncertainty description and H_P for provenance. A weighted bookkeeping cost can be proposed as

$$C_{\text{ledger}}(L; P) = \int_0^L c_P(x) \rho_P(x) dx, \quad (6.1)$$

provided c_P specifies the cost assigned to one registration and ρ_P corresponds to actual protocol requirements. Equation (6.1) is an accounting construction, not a statement that nature continuously performs those registrations. Correlations, redundancy and loss require a richer functional $\mathcal{C}[L, \Delta, H_P, U_P, \Sigma, P]$ if one wants the cost of a recoverable record.

This also separates geometrical extent from interactional extent. A metre gives a nominal distance. An observation gives a distribution of finite responses through time, aperture, material and reconstruction. To assert a relationship between the two requires an instrument model. The finite window needed to register a result is especially relevant when a classical argument uses an idealised instantaneous value or an infinitely sharp boundary. The model may retain that idealisation as a useful mathematical operation, while the FSM account records what was compressed by it.

Chapter 7

Chains, compression and remainders

A registration is rarely turned into a physical claim in one move. Let B_0 be the raw interaction record and let $B_{j+1} = T_j(B_j; \mathcal{L}_j)$ be a new bead obtained through a declared transformation: reading, calibration, fitting, assigning a unit, associating a line, or applying a model. Each T_j should retain an account of what information survives, what uncertainty is propagated and which sources of variation are removed or treated as irrelevant. The resulting finite symbolic trajectory is

$$B_0 \xrightarrow{T_0, \mathcal{L}_0} B_1 \xrightarrow{T_1, \mathcal{L}_1} \dots \xrightarrow{T_{n-1}, \mathcal{L}_{n-1}} B_n. \quad (7.1)$$

At each step κ_j records the distinction still relevant to the purpose of the comparison but no longer reconstructible from B_{j+1} alone. A transformed record may be useful despite its remainder. Multiple remainders should not automatically be added: their dimensions, correlations and locations in the chain may differ. When their values and covariances are known, a conventional uncertainty propagation or residual model can be applied; when they are not, the ledger should show the missing provenance rather than create an invented additive quantity.

In this sense the compact sign “1 m” can stand for a calibration history, an uncertainty budget, a unit definition and an intended resolution. It does not literally store all those data. Its ability to refer the reader back to them depends on an external practice, archive and licence. Decompression is an activity that uses those resources to construct a longer trajectory; it is not the release of hidden bytes physically contained in the printed characters. This distinction keeps the powerful metaphor of a bead representing many beads without confusing representation with material storage.

Chapter 8

A measured spectrum and an inferred distance

Spectroscopy provides a chain whose links can be separately examined: received light interacts with optics and detector; detector readings are calibrated; features are identified; positions become estimated wavelengths; observed positions are compared with reference lines; a redshift parameter is reported. For a chosen reference feature,

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{ref}}}{\lambda_{\text{ref}}} \quad (8.1)$$

is a ratio derived from spectral registrations and their calibration. Identifying a line, assigning a wavelength and estimating z have finite observational histories. They do not require a prior direct measurement of the source distance in parsecs. Distance is inferred through another chain that may draw on parallax, standard candles, other observations and a cosmological model. NASA's spectroscopy explanations describe line shifts; its astronomical accounts describe the further use of redshift in distance inference.¹

The analogy with sound Doppler measurements should likewise be decompressed with care. Wave mathematics provides useful relations across sound and electromagnetic cases; it does not prove that the media, trajectories, calibration procedures or distance inferences are identical. Nor is every astronomical redshift simply the ordinary Doppler effect of a source moving through a stationary laboratory. A productive FSM exercise is to place the local spectral comparison and the inferred distance side by side, marking the particular bridge at which a physical interpretation of z is introduced.

A large value of $N_{\text{cov}}(L, \Delta)$ for an inferred astronomical L is not itself evidence of an unmodelled spectral displacement. If the interactional account is to affect a predicted spectrum, a new rule must specify a coupling from a path dependent property to an observable, with units, sign, magnitude and dependence on line or environment. The ordinary model, instrument systematics and selection effects then provide the comparison. Without this bridge, representational cost remains a description of our records and chosen reconstruction.

¹NASA Webb, *Spectroscopy 101*, <https://science.nasa.gov/mission/webb/science-overview/science-explainers/spectroscopy-101-beyond-temperature-and-composition/>; NASA Hubble, *Cosmological Redshift*, <https://science.nasa.gov/mission/hubble/science/science-behind-the-discoveries/hubble-cosmological-redshift/>.

Chapter 9

What would make cost physically consequential?

We can write the missing bridge as a conditional hypothesis rather than hide it in a correction term. Let $\mathcal{I}_0$ describe a prepared interaction, $\mathcal{I}_L$ its registered outcome after an effective path L , and $Q_P(L)$ a physically defined path functional measurable or computable under protocol P . A candidate FSM model would have the form

$$\mathcal{I}_L = \mathcal{M}_0(\mathcal{I}_0, L, \theta) + \mathcal{B}(Q_P(L), \theta_B), \quad (9.1)$$

where $\mathcal{M}_0$ is a conventional baseline model and $\mathcal{B}$ a specified additional mapping. Before it becomes physics, Q_P must be independent of merely changing the scientist's notation. Replacing 10 pm with 20 pm in a ledger cannot alter a detector reading unless the altered scale changes an implemented protocol or stands for a independently evidenced property of the interaction. A measured dependence on actual path length, material, aperture, stabilisation, delay or registration conditions would supply a possible physical route; a count defined solely by L/Δ would not.

This yields two coherent levels of enquiry. On the symbolic level, assess what a formal equation compresses, compare histories and propagate uncertainty. On the physical level, propose a coupling with an operational variable and compare its prediction against measured residuals under the same registration rules. The bridge is explicit only when it maps a property of the physical interaction into a predicted change in a registered quantity. An unexplained improvement in fit is insufficient because calibration, nuisance parameters and correlated errors can also move a fitted residual.

The early k_{ma} can now be placed precisely: it signals that correspondence between independent finite trajectories requires inspection. It does not, without a new operational rule, tell us the size of a discrepancy in newtons. Equations (1.1) and (9.1) keep the representational remainder and possible physical addition separate, allowing experiments to decide whether the latter is needed for a particular phenomenon.

Chapter 10

A programme for classical results

The first programme is metrological. For one local length, keep the same physical object and make several records: a short unit inscription, a scalar estimate with a stated uncertainty, a sampled spatial profile, and a calibration history. Calculate the costs separately, repeat at several resolutions and determine whether the measured number of stable distinctions follows the proposed rule. Record response kernels, covariance and provenance. Compare a prototype-bar style chain with a modern optical realisation as different means of licensing “one metre”. This work tests the internal usefulness of the theory without presuming new dynamics.

A second programme is optical and local. In the proposed laser, pinhole and CCD experiment, vary actual source–aperture and aperture–detector separations, exposure, pixel aggregation and optical geometry. Register raw frames, dark and flat fields, spectral bandwidth, aperture dimensions, mechanical drift and the uncertainty of the measured ring radii. Construct delay embeddings only after retaining the order and timestamps of the frames; compare them with distributions obtained under deliberate reordering or fixed-condition controls. Fit an established diffraction model to the calibrated data, and then ask whether a reproducible remainder depends on *physical* path length or stabilisation protocol after known resolution and alignment effects have been accounted for. The resolution of an exported image can be changed without changing the light path: that manipulation is an important negative control for a claim that representational cost couples to optics.

The third programme examines equations with independently realised terms. For $F \mid ma$, list the force, mass and acceleration registrations and their calibration histories, then compare the constructed ma with the force record using a joint uncertainty model. Only if a repeatable remainder remains after the known interactional and instrumental terms have been represented should it be fitted as an additional physical component. Similar ledgers can be prepared for wave propagation, spectral line estimates and gravitational observables, with a declared licence for each bridge.

The fourth programme is astronomical and necessarily more demanding. Begin with spectra for which independent distance information exists. Analyse raw spectral registration, wavelength calibration and line assignment without inserting the distance into the redshift measurement. Construct the separate distance trajectory, including selection and model dependencies. Search for a proposed path functional only after expressing a falsifiable alternative to the standard predictions and showing that the result is stable across lines, objects, instruments and independent distance methods. The enormous nominal interval count of a parsec motivates examination of compression; by itself it gives no scale or sign for an astronomical correction.

Chapter 11

Propositions and open obligations

Under an expressly chosen resolution Δ and the coverage convention, the count $N_{\text{cov}}(L, \Delta)$ is finite for finite L and grows approximately linearly with extent. Under a stipulated explicit grid protocol with a constant assigned cost per interval, the grid cost grows in the same way. A short symbol can denote any of these extents while its surface length remains nearly constant. These statements follow from the chosen representation and need no new claim about the physics of light or gravity.

To obtain a theory of measured representations, each application must additionally declare the instrument, the distinction criterion, the usable Greene–Pascal trajectory, uncertainty, correlation and provenance. To obtain a theory with new physical predictions, it must further exhibit a physical path property, a rule coupling that property to an observable, and data that discriminate the rule from an appropriate conventional model. These are separate obligations; keeping them separate gives the central insight a form that can be tested rather than silently transferred across the equals sign.

The metre and the parsec thus expose two sides of the same achievement. A conventional unit and a small set of characters let us perform thought and calculation across extraordinary extents. When an enquiry demands the finer trajectory, the compressed sign instructs us where to begin reconstructing it, but the trajectory must be built through an actual licence and record. The practical FSM question is then: which finite distinctions were retained, which were released, and at what point would restoring one of them alter a physical prediction?

Afterword: Representation, Reconstruction, and the Inheritance of Models

The construction developed in this monograph should not be read as an attempt to discard the models that preceded it. Finite Symbolic Mechanics proceeds from a different position. Existing models are themselves finite symbolic constructions, developed under particular licences, commitments, historical circumstances, instruments, and mathematical practices. Their success lies partly in their capacity to compress long interactional and representational trajectories into compact symbolic forms that can subsequently be manipulated, communicated, and extended. The purpose of FSM is therefore not to erase those constructions, but to make their licensing visible and to ask what follows when different foundational commitments are adopted.

A model is not the termination of a trajectory. It is a temporarily stabilised compression that becomes a new working surface. Once that surface exists, subsequent work can reopen it, decompress particular parts, alter the commitments under which those parts are interpreted, and construct a finer representational trajectory. Scientific development can therefore be understood not simply as a succession of replacements, but as repeated cycles of registration, representation, compression, inheritance, and reconstruction.

A schematic form is

$$\mathcal{I} \longrightarrow \mathcal{R} \longrightarrow \mathcal{S} \longrightarrow \mathcal{C} \longrightarrow \mathcal{M} \longrightarrow \mathcal{L}' \longrightarrow \mathcal{R}',$$

where $\mathcal{I}$ denotes an interaction, $\mathcal{R}$ a registration, $\mathcal{S}$ a symbolic representation, $\mathcal{C}$ a compression, $\mathcal{M}$ a model, $\mathcal{L}'$ a subsequent licence or set of commitments, and $\mathcal{R}'$ a reconstruction developed under those commitments. The final operation does not imply that the earlier model was simply false. Rather, the earlier construction created a sufficiently stable symbolic surface for a later reconstruction to become possible.

This is particularly important when considering the mathematical notation used throughout the present work. A number of classical mathematical objects appear in the monograph, including ratios, summations, logarithms, covariance constructions, and, most conspicuously, the integral. Such notation is used because it provides an efficient and familiar symbolic carrier for relationships that would otherwise require lengthy exposition. Its use should not, however, be taken to imply that these mathematical objects are themselves admitted as primitive objects under the FSM licence.

For example, an expression such as

$$C_{\text{ledger}}(L; P) = \int_0^L c_P(x) \rho_P(x) dx$$

is useful as a compact conventional description of accumulated representational cost across an extent. The integral is here employed under an inherited classical mathematical licence. It acts as a symbolic surrogate for a construction that FSM would ultimately seek to express through finite registered extents, finite ordered interactions, finite uncertainties, finite overlap, and explicit provenance. The differential dx , the continuously indexed variable x , and the interval $[0, L]$ therefore should not be mistaken for foundational FSM objects merely because they occur within an FSM argument.

The same applies, in different degrees, to other familiar constructions. A finite summation may appear more compatible with finitist commitments than an integral, yet even

$$\sum_{i=1}^N c_i$$

performs an immediate compression. It discards the ordering of the contributing registrations unless that ordering is separately retained. It can discard overlap, provenance, uncertainty, interactional dependence, and the differences between trajectories that happen to yield the same scalar sum. A trajectory such as

$$R_1 \rightarrow R_2 \rightarrow R_3 \rightarrow \cdots \rightarrow R_N$$

contains representational structure that is not preserved merely by replacing it with a number.

A more native FSM object would therefore begin with the trajectory itself. Each registered extent may carry a structured set of finite properties,

$$R_i = (\Delta_i, \rho_i, \omega_i, U_i, H_i, c_i),$$

where Δ_i denotes finite extent, ρ_i interactional density, ω_i overlap, U_i uncertainty, H_i provenance, and c_i some locally assigned representational cost under the relevant licence. The trajectory

$$\mathcal{T}_P = R_1 \rightarrow R_2 \rightarrow \cdots \rightarrow R_N$$

is then prior to any scalar compression of that trajectory.

Only after the finite trajectory has been constructed should an operation be licensed that compresses it into a simpler descriptor. One may therefore distinguish the trajectory from a subsequent compression rule,

$$\mathcal{K}_{\mathcal{L}} : \mathcal{T}_P \longrightarrow C_P,$$

where $\mathcal{K}_{\mathcal{L}}$ indicates that a particular compression has been performed under licence $\mathcal{L}$. The resulting scalar C_P may be computationally useful, but it should not be confused with the finite interactional trajectory from which it was derived.

This distinction reveals something important about mathematical notation itself. The integral sign is not merely a mathematical operator. It is also an example of symbolic compression. The symbol

$$\int$$

carries within it a long inherited mathematical trajectory involving accumulation, partition, limiting procedures, continuity, measure, and the conventions through which such constructions became admissible within classical mathematics. In this sense, the mathematical notation employed to describe physical compression is itself a compressed symbolic construction.

There are therefore at least two simultaneous levels of compression within an equation of physical theory. The first concerns the physical or experimental trajectory:

interaction $\longrightarrow$ compressed physical symbol.

The second concerns the mathematical operations performed upon those symbols:

symbolic operational trajectory $\longrightarrow$ compressed mathematical object.

An expression containing a distance, an integral, a fitted parameter, and a named physical object may consequently contain several nested layers of compression. The equation remains useful precisely because those layers have been made symbolically inexpensive. FSM does not deny that utility. It asks that the inherited compression should remain recoverable in principle through provenance and licensing.

This provides a general way to understand scientific reconstruction. Earlier scientific workers do not transmit their original interactions directly to later generations. They transmit instruments, inscriptions, tables, equations, diagrams, texts, conventions, and named objects. These are stabilised symbolic beads carrying compressed histories. A later investigator receives those beads, interprets them under a current licence, and constructs further trajectories from them.

Newton did not receive Galileo's original interactional world. Maxwell did not receive Faraday's interactions directly. Modern investigators do not receive Maxwell's registrations. What passes between generations is an inherited symbolic structure containing compressed traces of those earlier trajectories. Scientific objects are therefore inherited compressions before they become objects of new construction.

This applies not merely to numerical quantities but to many of the central objects of scientific language. A metre, a parsec, a field, a photon, a wavelength, a spectral line, an integral, a particle, and a wave all possess histories of construction, stabilisation, use, and compression. Once established, such terms become inexpensive symbolic beads. They can be placed into new chains without repeatedly reproducing the trajectories that first licensed them.

This symbolic economy is one of the great strengths of science. Without compression, every physical argument would require the reconstruction of its entire historical and experimental ancestry. Yet the same economy creates a foundational risk. When the history of a bead is no longer visible, the compressed representation can come to be treated as though it were identical with the interaction from which it ultimately arose.

Provenance therefore becomes foundational rather than merely documentary. It provides a route by which a compressed symbolic object can, where necessary, be reopened. Reconstruction becomes possible because the bead retains, either explicitly or through recoverable history, a trajectory leading back toward the interactions and licences from which it was formed.

Scientific development may consequently be represented as a recursive sequence of reconstruction and compression,

$$\mathcal{R}_0 \longrightarrow \mathcal{C}_1(\mathcal{R}_0) \longrightarrow \mathcal{R}_1 \longrightarrow \mathcal{C}_2(\mathcal{R}_1) \longrightarrow \mathcal{R}_2 \longrightarrow \cdots,$$

where each compression produces a new stable symbolic surface and each reconstruction reopens some part of that surface under a new or modified licence.

Within this framework, an earlier model need not be rejected merely because a later licence exposes structure that it did not retain. Its compression may remain exceptionally effective within its intended range. Reconstruction becomes necessary only where the commitments of the new licence require distinctions that the earlier representation removed.

This provides a useful perspective on the classical mathematical objects retained in the present monograph. They should be regarded as inherited symbolic machinery used provisionally to make the present construction possible. Where they remain useful, they can continue to be employed. Where their foundational commitments conflict with those of FSM, they can later be reopened and reconstructed.

The path from the classical integral to a finite ordered interactional trajectory is therefore not an embarrassment or inconsistency. It is an example of the process being described. The classical object enables the present argument to be constructed sufficiently far that its own compressed assumptions can subsequently be examined. In this sense the borrowed notation provides the working surface upon which a finer FSM object can later be built.

The earlier development of the residual term in Finite Mechanics can also be viewed through this lens. An expression such as

$$F \mid ma + k_{ma}$$

originally preserved a non-zero component that resisted complete symbolic closure under the commitment to finite physical quantities. At that earlier stage, k_{ma} appeared to demand interpretation as a physical thing, correction, error, or additional force because the surrounding scientific language strongly encouraged every mathematical term to be treated as naming a separable object.

The later development of finite symbolic trajectories, provenance, compression, interactional identity, licensing, and representational cost permits a different reading. The residual may be understood not primarily as another physical entity but as a mark of incomplete representational identity between independently constructed finite trajectories. It leaves visible something that exact equality suppresses.

In that sense, the residual can be seen as an early opening in the symbolic closure of the model. The later Theory of Finite Representation supplies a vocabulary for understanding why such an opening may be required. A finite physical representation does not become identical to another finite physical representation merely because their mathematical surrogates can be declared equal within a classical formal system.

The same methodological principle applies throughout the present programme. A classical model can be retained, its licence identified, its symbolic compression exposed, and its trajectory reconstructed only where required. FSM therefore does not begin by demanding the removal of existing mathematics or physics. It begins by asking what has been licensed, what has been compressed, what has been inherited, and what additional distinctions become necessary once finite representation itself is taken as a foundational commitment.

The resulting scientific trajectory is therefore neither simple preservation nor simple replacement. It is reconstruction through inheritance.

A model becomes a bead. The bead becomes a working object. The working object enters a new trajectory. A later licence reopens the bead. A finer construction becomes

possible.

The process can continue.

Scientific objects are inherited compressions before they become objects of new construction.

The present monograph should therefore be read as one stabilised point within that continuing trajectory. Its use of classical formal notation is deliberate and provisional. Those objects permit the argument to be carried forward, while their own representational commitments remain available for subsequent examination and reconstruction within the wider development of Finite Symbolic Mechanics.

Terminological note

In this working monograph *Alphonic scale* denotes a declared, provisionally distinguishable finite scale; *Greene–Pascal condition* denotes a criterion for a usable finite symbolic trajectory. Neither term is an SI unit. *Coverage count* is a chosen resolution count; *cost* requires an indicated component, encoding and task. The vertical bar denotes a declared FSM correspondence between registrations or constructions, and κ a comparison remainder whose mathematical type must be supplied by the application. *Licence* refers to the commitments that make a translation or comparison admissible; it does not claim that the underlying formal mathematics is invalid.